CYBERSECURITY & INFRASTRUCTURE

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Academic Lab Portfolio

Practical work in network defense, routing, cloud infrastructure, and local AI. The reports combine configuration records, screenshots, and troubleshooting lessons from hands-on academic exercises.

01 / NETWORK SECURITY

Layer 2 Attacks and Mitigations

MAC flooding, ARP spoofing, and DHCP starvation.

Evidence: Packet captures, switch controls, and recovery troubleshooting.

02 / NETWORK ROUTING

Multi-Area OSPF Routing

Cisco router configuration and adjacency troubleshooting.

Evidence: Annotated topology, device configurations, and neighbor troubleshooting.

03 / NETWORK ROUTING

BGP and Route Redistribution

Interconnecting OSPF, EIGRP, and IS-IS routing domains.

Evidence: Addressing tables, command references, configurations, and route output.

04 / CLOUD INFRASTRUCTURE

AWS Foundations: IAM, VPC, and EC2

Identity permissions, network segmentation, and compute lifecycle.

Evidence: Permission tests, subnet routing, web-server output, and instance monitoring.

05 / CLOUD INFRASTRUCTURE

AWS Storage, Databases, and High Availability

EBS, Amazon RDS, load balancing, and Auto Scaling.

Evidence: Database connectivity, target health checks, and scaling observations.

06 / LOCAL AI

Local AI with Ollama and Open WebUI

Windows setup, container deployment, and browser access.

Evidence: Model interaction, terminal output, container views, and a browser interface.

Full reports are provided in PDF and editable Word format. Configuration excerpts reflect the original lab environments; corrected reference examples are included in the affected reports.